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On Reduction Rules, Meaning-as-use, and Proof-theoretic Semantics

Abstract. The intention here is that of giving a formal underpinning to the idea of 'meaning-is-use' which, even if based on proofs, it is rather different from proof-theoretic semantics as in the Dummett–Prawitz tradition. Instead, it is based on the idea that the meaning of logical constants are given by the explanation of immediate consequences, which in formalistic terms means the effect of elimination rules on the result of introduction rules, i.e. the so-called reduction rules. For that we suggest an extension to the Curry–Howard interpretation which draws on the idea of labelled deduction, and brings back Frege's device of variable-abstraction to operate on the labels (i.e., proof-terms) alongside formulas of predicate logic.

Keywords: natural deduction; reduction rules; proof-terms; proof-theoretic semantics; game-theoretic semantics; dialogue semantics

1. Proof-theoretic semantics

In a recent special issue of *Synthese*, edited by R. Kahle & P. Schroeder-Heister (Volume 148, Number 3 / February, 2006), the possibility of a "proof-theoretic semantics" is put forward in the Editorial:

> According to the model-theoretic view, which still prevails in logic, semantics is primarily denotational. Meanings are denotations of linguistic entities. The denotations of individual expressions are objects, those of predicate signs are sets, and those of sentences are truth values. The meaning of an atomic sentence is determined by the meanings of the individual and predicate expressions this sentence is composed of, and the meaning of a complex sentence is determined by the meanings of its constituents. A consequence is logically valid if it transmits truth from its premisses to its conclusion, with respect to all interpretations. Proof systems are shown to be correct by demonstrating that the consequences they generate are logically valid. This basic conception also underlies most alternative logics such as

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> intensional or partial logics. In these logics, the notion of a model is more involved than in the classical case, but the view of proofs as entities which are semantically dependent on denotational meanings remains unchanged.
> Proof-theoretic semantics proceeds the other way round, assigning proofs or deductions an autonomous semantic role from the very onset, rather than explaining this role in terms of truth transmission. In proof-theoretic semantics, proofs are not merely treated as syntactic objects as in Hilbert's formalist philosophy of mathematics, but as entities in terms of which meaning and logical consequence can be explained.

Still in the same Editorial the origins of the term is explained as:

> The programme of proof-theoretic semantics can be traced back to Gentzen (1934). Seminal papers by Tait, Martin-Löf, Girard and Prawitz were published in 1967 and 1971. An explicit formulation of a semantic validity notion for generalized deductions with respect to arbitrary justifications was given by Prawitz (1973). Much of the philosophical groundwork for proof-theoretic semantics was laid by Dummett from the 1970s on, culminating in Dummett (1991). Martin-Löf's type theory, whose philosophical foundation is proof-theoretic semantics, became a full-fledged theory in the 1970s as well (see Martin-Löf 1975, 1982). The term "proof-theoretic semantics" was proposed by the second editor in a lecture in Stockholm in 1987.

(Here it must be mentioned that already in 1968 Franz von Kutschera [46] used the term 'Gentzen Semantik' to refer to the idea of defining the meaning of the logical operations by the left and right introduction rules of a sequent calculus.[1])

Gentzen's analysis of logical deduction gave yet one more argument in favor of the view that the proof conditions determine the meaning of a proposition:

> The introductions represent, as it were, the 'definitions' of the symbols concerned, and the eliminations are no more, in the final analysis, than the consequences of these definitions.[2]

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[1]Thanks are due to Heinrich Wansing for this historical point, which is discussed in more detail in his own paper on the idea of a proof-theoretic semantics [48].

[2][25], p. 80 of the English translation in [44].

In a series of lectures entitled 'On the meaning of logical constants and the justification of the logical laws' (1985), Martin-Löf presents philosophical explanations concerning the rôle of the definition of direct/canonical proofs in the proof-based account of meaning:

> The intuitionists explain the notion of proposition, not by saying that a proposition is the expression of its truth conditions, but rather by saying, in Heyting's words, that a proposition expresses an expectation or an intention, and you may ask, An expectation or an intention of what? The answer is that it is an expectation or an intention of a proof of that proposition. And Kolmogorov phrased essentially the same explanation by saying that a proposition expresses a problem or task (Ger. *Aufgabe*). Soon afterwards, there appeared yet another explanation, namely, the one given by Gentzen, who suggested that the introduction rules for the logical constants ought to be considered as so to say the definitions of the constants in question, that is, as what gives the constants in question their meaning. What I would like to make clear is that these four seemingly different explanations actually all amount to the same, that is, they are not only compatible with each other but they are just different ways of phrasing one and the same explanation.[3]

The idea is further pursued by one of the guest editors in his own contribution to the aforementioned volume, which purports to call Gentzen's remarks 'the most specific root' of the proposed account of proof-theoretic semantics:

> Proof-theoretic semantics has several roots, the most specific one being Gentzen's (1934) remarks that the introduction rules in his calculus of natural deduction define the meanings of logical constants while the elimination rules can be obtained as a consequence of this definition. More broadly, it belongs to the tradition according to which the meaning of a term has to be explained by reference to the way it is used in our language.[4]

As a matter of fact, the idea that the meaning of a proposition is given by what counts as a proof of it was put forward by Dummett and Prawitz in the early 1970's, and the whole programme became known as the *Dummett-Prawitz argument*. First Dummett:

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[3] [33, p. 410].

[4] [43, p. 525f].

> the meaning of each [logical] constant is to be given by specifying, for any sentence in which that constant is the main operator, what is to count as a proof of that sentence, it being assumed that we already know what is to count as a proof of any of the constituents.[5]

Then Prawitz:

> As pointed out by Dummett, this whole way of arguing with its stress on communication and the role of the language of mathematics is inspired by ideas of Wittgenstein and is very different from Brouwer's rather solipsistic view of mathematics as a languageless activity. Nevertheless, as it seems, it constitutes the best possible argument for some of Brouwer's conclusions. (...)
> I have furthermore argued that the rejection of the platonistic theory of meaning depends, in order to be conclusive, on the development of an adequate theory of meaning along the lines suggested in the above discussion of the principles concerning meaning and use. Even if such a Wittgensteinian theory did not lead to the rejection of classical logic, it would be of great interest in itself.[6]

So strong was Prawitz' confidence in the argument that he proposed to give a kind of 'technical' counterpart to Gentzen's dictum that the so-called *introduction* (proof) rules give the meaning to the logical constants. A general schema for obtaining the elimination rules from the introduction rules of any logical constant was proposed in a paper published in 1978:

> Gentzen suggested that "it should be possible to display the elimination rules as unique functions of the corresponding introduction rules on the basis of certain requirements." One has indeed a strong feeling that the elimination rules are obtained from the corresponding introduction in a uniform way. For the introduction rules of the schematic type, we can easily describe how the elimination rules are obtained uniformly from the corresponding introduction rule.[7]

As a result, a set of 'generalized elimination rules', following the pattern of the elimination of disjunction, was put forward. The resulting rule of elimination of implication, however, turned out to be rather different from

[5] [20, p. 12].
[6] [37, p. 18].
[7] [38, p. 36f].

the usual *modus ponens*, and seemed to carry a certain degree of circularity, as in:

$$\frac{A \to B \qquad \left[\begin{array}{c} A \\ \vdots \\ B \end{array}\right] \\ \qquad\qquad C}{C}$$

meaning that, if, by assuming that one can get B from A one is able to obtain C, then one can conclude C from $A \to B$ no longer relying on the extra assumption that B is obtainable from A.

Note that the rule involves an assumption which is itself a hypothetical deduction, and which is dicharged by the application of the rule. A further development by P. Schroeder-Heister called 'A Natural Extension to Natural Deduction' [42] argues for the adoption of such a general account of hypothetical reasoning where rules themselves may be introduced as assumptions (and later discharged).

In spite of Schroeder-Heister's proposed way out of the circularity shown in Prawitz' general schema for obtaining/deriving the elimination rules from the introduction rules, there was an attempt at a revision of the whole enterprise along the lines of the proposal that the elimination rules should be seen, not as *derivable from*, but rather as *justified by* the introduction rules. In his 1991 book *The Logical Basis of Metaphysics*, Dummett says:

> Gerhard Gentzen, who, by inventing both natural deduction and the sequent calculus, first taught us how logic should be formalised, gave a hint how to do this, remarking without elaboration that 'an introduction rule gives, so to say, a definition of the constant in question', by which he meant that it fixes its meaning, and that the elimination rule is, in the final analysis, no more than a consequence of this definition. (...) Plainly, the elimination rules are not consequences of the introduction rules in the straightforward sense of being derivable from them; Gentzen must therefore have had in mind some more powerful means of drawing consequences. He was also implicitly claiming that the introduction rules are, in our terminology, self-justifying.
> (...)
> It is confusing to speak of the elimination rules as *consequences* of the introduction rules: it is better to speak of them as being *justified* by reference to them.[8]

[8] [21, p. 251f].

He goes further to suggest an opposing view, namely the one where it is the elimination rule which fixes the meaning of the logical constants:

> Intuitively, Gentzen's suggestion that the introduction rules be viewed as fixing the meanings of the logical constants has no more force than the converse suggestion, that they are fixed by the elimination rules; intuitive plausibility oscillates between these opposing suggestions as we move from one logical constant to another. Per Martin-Löf has, indeed, constructed an entire meaning-theory for the language of mathematics on the basis of the assumption that it is the elimination rules that determine meaning. The underlying idea is that the content of a statement is what you can *do* with it if you accept it—what difference learning that it is true will, or at least may, make to you. This is, of course, the guiding idea of a pragmatist meaning-theory. When applied to the logical constants, the immediate consequences of any logically complex statement are taken as drawn by means of an application of one of the relevant elimination rules.[9]

This seems to be a clear departure, on the part of Dummett, from the shared vision that the Gentzen's dictum was the ultimate justification for a theory of meaning for the language of intuitionistic mathematics. An indeed, in a paper published in 1998, Dummett clearly steps out of the joint programme with Prawitz, this time acknowledging explicitly that Prawitz' perspective is different from (and rival to) Wittgenstein's own view on the connections between meaning and use:

> Prawitz's view is a genuine rival to Wittgenstein's only if we do understand the notion of truth as having metaphysical resonance. If we do, then we need to think very hard about the basis of the constructivist view of meaning. I presented it as deduced from a principle and an ancillary thesis. The principle was that the meaning of any given form of statement is encapsulated in what we learn when we learn to use statements of that form; and this principle must surely be correct. How could we come to attach a meaning to any such statements if a grasp of it were not intrinsic to the use we make of them? The ancillary thesis, as I stated it, was crude: it was to the effect that all that we learn, when we learn the use of a given form of statement, was what constitutes the verification of a statement of that form, or what establishes its truth and hence conclusively warrants an assertion of it. I already acknowledged that, as applied to

[9] Ibid., p. 280.

> empirical statements, this is too restrictive; grounds that fall short of being conclusive may also be true in virtue of a past, but no longer present, possibility of verifying it presents a different reason for qualifying the ancillary thesis. I regret that I do not feel able at present to offer a clear formulation of the necessary qualification; but it surely involves a fairly substantial concession to realism.[10]

In a paper published in 1998 Prawitz remains confident that Gentzen's claim should be understood in the sense that the elimination rules are *justified* on the basis of the meaning given to the logical constants by the introduction rules:

> (...) This may sound as a formalist position: the meaning of the logical constants are given by inference rules that are just taken for granted, in other words, their validity is not thought of as depending on anything. However, Gentzen's point was that only inference rules of a certain kind, that is, the introduction rules, were given as valid in the sense that they just state what we mean by the sentences that occur as conclusions of the inferences in question. Other inference rules were to be justified on the basis of the meaning given to the logical constants by the introduction rules.[11]

> A way of formulating Gentzen's idea more systematically in semantic terms is to start from the idea that the meaning of a statement is determined by what counts as a proof of the statement, an idea formulated already before Gentzen by Heyting in his account of intuitionism.[12]

> That the meaning of a statement is determined by what counts as its canonical proof or verification is thus the essence of the verificationism that I have in mind.[13]

But in a paper published in 2002, Prawitz reveals a certain degree of flexibility:

> As Michael Dummett has put it, the rules that govern the use of expressions may be divided into two kinds: the ones that tell on

[10] [22, p. 137f].
[11] [39, p. 42].
[12] Ibid., p. 44.
[13] Ibid., p. 45.

> what conditions it is appropriate to utter them, and the ones that concern the appropriate reactions to the utterance. In the case of assertive sentences, the latter rules concern the expectations we have when trusting an assertion, or, in other words, the conclusions that can be rightly inferred from the asserted sentence. The latter aspect is an aspect of the use of sentences in another sense of "use" again, namely what such sentences are used for, what they are good for. With Dummett we may call a theory of meaning pragmatic when it is based on this other aspect of the use of expressions. What are the prospects for such a theory? Is it more promising?
> (...)
> No one has really tried to construe a comprehensive semantics built on the idea that the meaning of a sentence is given by what counts as direct consequences of the sentence. It seems easy enough to work this idea for the language of predicate logic. Unfortunately, it seems less easy to do the same for mathematics.[14]

The prospect for an alternative account for the relation between proofs and meaning which departs from the original formulation of the so-called 'Dummett-Prawitz argument' is also mentioned in a more recent publication:

> The term proof-theoretic semantics would have sounded like a *contradictio in adjecto* to most logicians and philosophers half a century ago, when proof theory was looked upon as a part of syntax, and model theory was seen as the adequate tool for semantics. Michael Dummett is one of the earliest and strongest critics of the idea that meaning could fruitfully be approached via model theory, the objection being that the concept of meaning arrived at by model theory is not easily connected with our speech behaviour so as to elucidate the phenomenon of language. (...)
> I shall leave the prospects of rightly connecting the meanings of expressions with our use of them within a theory of meaning developed along these lines, and shall instead review some approaches to meaning that are based on how we use sentences in proofs. One advantage of such an approach is that from the beginning meaning is connected with aspects of linguistic use.[15]

Indeed, since Dummett's attempt to give a philosophical justification of intuitionistic logic from a 'linguistic' perspective, drawing on Wittgenstein's

[14] [40, p. 92].
[15] [41, p. 507f].

dictum 'meaning-is-use', the idea of giving an account of the semantics of the language of mathematics which purports to explain the relation between proof, inference rule, and meaning has motivated a number of developments, most, if not all, in line with Gentzen's suggestion that the meaning of a logical constant is fully specified by the corresponding conditions for assertion. In a recent article, Schroeder-Heister says:

> Proof-theoretic semantics is an alternative to truth-condition semantics. It is based on the fundamental assumption that the central notion in terms of which meanings can be assigned to expressions of our language, in particular to logical constants, is that of *proof* rather than *truth*. In this sense proof-theoretic semantics is inherently inferential in spirit, as it is the inferential activity of human beings which manifests itself in proofs.
>
> Proof-theoretic semantics has several roots, the most specific one being Gentzen's (1934) remarks that the introduction rules in his calculus of natural deduction define the meanings of logical constants, while the elimination rules can be obtained as a consequence of this definition. More broadly, it is part of the tradition according to which the meaning of a term should be explained by reference to the way it is *used* in our language.
>
> Although the "*meaning as use*" approach has been quite prominent for half a century now and has provided one of the cornerstones of the philosophy of language, in particular of ordinary language philosophy, it has never prevailed in the formal semantics of artificial and natural languages. In formal semantics, the denotational approach, which starts with interpretations of singular terms and predicates, then fixes the meaning of sentences in termsof truth conditions, and finally defines logical consequence as truth preservation under all interpretations, has always been dominant. The main reason for this, as I see it, is the fact that from the very beginning, denotational semantics received an authoritative rendering in Tarski's (1933) theory of truth, which combined philosophical claims with a sophisticated technical exposition and, at the same time, laid the ground for model theory as a mathematical discipline. Compared to this development, the "meaning as use" idea was a slogan supported by strong philosophical arguments, but without much formal underpinning.[16]

[16] [43, p. 525f]. Here it must be mentioned some of the early formal work on functional completeness in proof-theoretic semantics, the earliest one being two papers by von Kutschera [46, 47].

It is exactly with the intention of giving a formal underpinning to the idead of 'meaning-is-use' that we develop below an alternative account of proof-theoretic semantics based on the idea that the meaning of logical constants are given by the explanation of immediate consequences, which in formalistic terms means the effect of elimination rules on the result of introduction rules, i.e. the so-called reduction rules.[17]

2. Meaning, use, and consequences

Further to the connections between *meaning* and *use*, it seems useful to consider the (explanation of the immediate) *consequences* one is allowed to draw from a proposition as something directly related to its *meaning/use*. And indeed, Wittgenstein's references to the connections between *meaning* and the *consequences*, as well as between *use* and *consequences* are sometimes as explicit as his celebrated 'definition' of *meaning* as *use* given in the *Investigations*. In this section we attempt to collect some of these references, discussing how an intuitive basis for the construction of a more convincing proof-theoretic semantics (than, say, assertability conditions semantics) for the mathematical language can arise out of this connection *meaning/use/*(explanation of the immediate) *consequences*.

As a matter of fact, there is more to the relation between meaning, use, and the (explanation of the immediate) consequences of an assertion. The profound insights into the connections between *meaning* and *use*, *usefulness*, *purpose* and *function* provided by Wittgenstein's later writings, such as the remark: "Meaning, function, purpose, usefulness — interconnected concepts."[18] serve as a remarkably intuitive basis on which a proof-theoretic semantics for the mathematical language can be constructed. Not just another semantics but one that proves more convincing than many 'verificationistic' semantics. It suggests that in addition to prescribing that 'the meaning of a proposition is determined by its method of verification', no canonical value or mathematical construction is said to have meaning in itself, assertability conditions do not suffice, and no dialogical notions such

[17]This is part of an ongoing project which started with the reformulation of Martin-Löf's type theory into *Meaning-as-Use Type Theory* in [8] and [18], and whose main results have been published in a series of papers: [4], [5], [6], [7], [18], [9], [19], [10], [24], [11], [17], [12], [14], [15], [16] [17], [3], [2], [13]. Parts of the present paper will have appeared before, even if in a slightly modified form or context.

It has recently been brought to our attention that our proposal on the relationship between reduction rules and meaning has been analysed by P. Contu [1] and E. Moriconi [34].

[18]§291, p. 41^{e} of the *Last Writings on the Philosophy of Psychology*, Vol. I.

as game or dialogue need to be introduced. Accepting the explanation of the (immediate) *consequences* one is allowed to draw from a proposition as the explanation of its *meaning*, the explanation of how it is to be *used*, one may reasonably take that *meaning*, *use* and *consequences* are indeed interconnected concepts. Further to the connections between *meaning* and *use*, it seems useful to consider the *consequences* one is allowed to draw from a proposition as something directly related to its *meaning/use*. The resulting 'picture' is a triangle-like diagram with vertices on meaning-use-consequences, which highlights the rôle of the explanation of the *consequences* one can draw from a proposition for the settlement of its meaning as a fundamental ingredient of Wittgenstein's development of his celebrated semantical paradigm *meaning-is-use*. The main point here is to demonstrate how a particular formal counterpart to the explanation of (immediate) consequences, namely the proof-theoretic device of normalisation, can be used to explain some fundamental concepts of the logical foundations of computation. (E.g. the λ-calculus and the rule for explaining the *meaning/use/usefulness/consequences* of a λ-term: the β-rule of normalisation.)

Despite the technical presentation given at the Appendix, the chief motivation arises from an attempt to understand the question as to what extent Wittgenstein's philosophies might be seen as natural developments of a (roughly) uniform intuition concerning the relationship between philosophy and critique of language, and its implications to the foundations of mathematical logic. Recent studies based on the catalogued *Nachlass* have helped to uncover many misconceptions generated by the rather fragmented character of Wittgenstein's philosophical writings. In the light of the recent findings, it seems reasonable to argue, for example, that the logical atomism of the *Tractatus* was rejected by Wittgenstein in so far as it assumed that a name denoted an object, and the object was its 'meaning' (i.e. *denotation*). The opening lines of the *Investigations* (and beginning of: MS 111, TS 213, *Blue Book*, *Brown Book*) reveals one of the main criticisms against the *Tractatus*' chemical-like analysis: the assumption that names denote objects which are their meanings. In the later phase, whilst that assumption is dropped, the atomism idea is retained, leaving a sort of 'functional atomism', where words are taken to be individually identifiable devices with a meaning (function, purpose, usefulness) in the calculus of language. It can be seen as a revision of his early ('official') *strict atomism* into a kind of *manageable holism*, which accounts for both the continuity (i.e. atomism, with a 'functional' ('manageably holistic') ingredient, as D. Pears confirms: "the idea that a sentence can only be explained by their places in systems ...

is one of the central points of the picture theory"[19] and "Everyone is aware of the holistic character of Wittgenstein's later philosophy, but it is not so well known that it was already beginning to establish itself in the *Tractatus*."[20]) and the discontinuity (i.e. 'meaning as denotation' vs. 'meaning as use') of his main philosophies.

As an attempt to provide a concrete example of how such a functional atomism might be used to construct a framework where the logical connectives have a meaning (function, purpose, usefulness) in the calculus of language which is made clear by the explanation of the (immediate) consequences one is supposed to draw from the corresponding proposition, we discuss the so-called 'functional interpretation' of logics. The (Curry–Howard) functional interpretation of logical connectives is an 'enriched' system of Natural Deduction, in the sense that terms representing proof-constructions are carried alongside the formulas. It can be used to formulate various logical calculi in an operational manner, where the way one handles assumptions determines the logic. It is naturally atomistic, it is verificationist, and it is functional in the sense that the main semantical device is that of normalisation of proofs (not assertability conditions), finding a parallel in the dialogue/game-like semantics of Lorenzen (and Hintikka): canonical terms (*introduction*) stand for the 'Assertion' (resp. 'Myself') moves, and non-canonical terms (*elimination*) stand for the 'Attack' (resp. 'Nature') moves. It seems to comply with Wittgenstein's 'functional atomism': it accounts for the early atomism (already with holistic tendencies according to D. Pears [36]) in a functional fashion, abiding by 'words are tools' and 'language is a calculus' (which, S. Hilmy verifies, "was already in place in early 1930s, and remained at least until 1946"[21]), and is compatible with "asking whether and how a proposition can be verified is only a particular way of asking "How d'you mean?"" (*Investigations*, Part I, §353).

It seems as though signs of Wittgenstein's attempt to make the connections between explaining the *meaning* of a proposition, demonstrating

[19] footnote 34, p. 121 of [35].

[20] Opening lines of [36].

[21] Cf. Hilmy [27]. Hilmy's extensive study of the *Nachlass* has greatly helped removing classical misconceptions such as Hintikka's claim that "Wittgenstein in the *Philosophical Investigations* almost completely gave up the calculus analogy.", p. 14 of [30].

Hilmy points out that even in the *Investigations* one finds the use of the calculus/game paradigm to the understanding of language, such as "im Laufe des Kalküls" (Part I, §559) (in the English translation: "in operating with the word"), and "it plays a different part in the calculus" (footnote of p. 14). Hilmy also quotes from a late (1946) unpublished manuscript (MS 130) "this sentence has use in the calculus of language".

how to *use* it, and showing the (immediate) *consequences* one is allowed to draw from it, can be found in many different presentations. One finds many relevant remarks on the topic scattered among his writings, irrespective of phases. (The earliest reference to that particular aspect is perhaps in a letter sent to Russell in 1912, as we have previously pointed out, and later reinforce (section 6). In the later writings, however, the remarks of that nature become much more frequent, and significantly more incisive than those of the early phase. In a series of *Lectures on the Foundations of Mathematics* given in Cambridge in 1939, for example, he explains how the meaning of a proposition can be changed by changing the way it is used, by changing the consequences that can be drawn from it. The core of the discussion can be summarised in the following:

> *Wittgenstein:* Yes; and let us take another example; the use of "all". "If all the chairs in this room were bought at Eaden Lilley's then this one was. $(x).fx$ entails fa." Suppose I ask, "Are you sure fa follows from $(x).fx$? Can we assume that it does not follow? What would go wrong if we did assume that?"
> *Wisdom:* One reply which might be given is that it is impossible to make such an assumption.
> *Wittgenstein:* Yes. But let us look into this, because such things as "Let us assume that $(x).fx$ does not entail fa" have been said. Now the reply you suggested did not mean that it is psychologically impossible to assume that; for if it did, one might say that although Wisdom cannot do it yet perhaps other people can.
> In what way is it impossible to assume $(x).fx$ does not entail fa?
> *Wisdom:* Isn't the assumption like saying "Couldn't we have a zebra without stripes?"
> *Wittgenstein:* Yes. It would be said that the meaning of '$(x).fx$' had been changed.
> What then would go wrong if someone assumes that $(x).fx$ does not entail fa? I would say that all I am assuming is a different use of "all", and there is *nothing* wrong in this.
> If I stick to saying that the meaning is given by the use, then I cannot use an expression in a different way without changing the meaning. But it is then misleading to say, "The expression *must* have a different meaning if used differently." It is merely that it *has* a different meaning—the different use *is* the different meaning.
> And if one says, "If one assumes fa does not follow from $(x).fx$, one must use $(x).fx$ in a different way"—we reply, in assuming this one

does it in a different way.—But if we make this assumption, nothing goes wrong.[22]

From the above, it would not be unreasonable to say, for example, that not only the meaning of a proposition changes when the way it is used changes, when what can be deduced from it changes, but that the conclusions one can draw from a proposition *determine* its meaning. This view becomes much sharper and the evidences for it much clearer in Wittgenstein's very late writings, and that is what we shall attempt to demonstrate below. (And it is for that reason that we attempt to provide historical information such as dates, typescripts and manuscripts numbers whenever appropriate, etc.).

What we do here in trying to relate those three concepts of *meaning*, *use* and *consequences* by means of Wittgenstein's (mainly later) writings is in effect an attempt to show that a semantical standpoint based on the paradigm 'meaning is use' can be not just practically useful, as far as a basis for operational semantics of formalised languages is concerned, but it can also be very natural for a semantical standpoint supporting logic and computing science as a whole where meaning and rules have a close interplay. In order to define a logical sign by means of deduction/computation rules what one does, besides giving the 'grammatical' rules describing the assertability conditions, is essentially to demonstrate what can be obtained from the 'term' having the sign as its major one after *deducing*, *eliminating* the logical sign.

3. Meaning and use

There has been a number of different readings of Wittgenstein's 'definition' of meaning as given in "the meaning of a word is its use in the language".[23] Much has already been said on the different interpretations one can give to such a 'definition'. *Use* being such a vague concept as it is in ordinary language, enough room is left for distinct and perhaps uncongenial readings of Wittgenstein's semantical paradigm. There are readings discussed in the literature which lend too vague a character to Wittgenstein's use-based semantical paradigm, some attempting to discuss use in the sense of 'usage', some in the sense of 'practice', etc. The general picture which remains is that of vagueness, usually in the least positive sense. As a general rule, one's first reaction to the adoption of a use-based semantics in the analysis of formalised languages is that of scepticism with respect to the feasibility

[22] p. 192 of C. Diamond's *Wittgenstein's Lectures on the Foundations of Mathematics, Cambridge, 1939*.

[23] §43, Part I, p. 20^{e} of the *Philosophical Investigations*.

of the enterprise. One is induced to think that the enterprise may turn out to be unworkable, because when describing the use of a word or sentence one has to refer to the whole context within which it is embedded, and therefore, the holistic character of language is definitely 'unmanageable'. And from this, one is almost inevitably led to conclude that, according to Wittgenstein, 'language is really unanalysable' (*not* Wittgenstein's own words!). One might then justify the arrival at such a negative conclusion by saying that if one has to deal with meaning as functionality, purpose, usefulness, in a calculus/game/system of logical signs, without having the support of some 'universe' of values (cf. interpretations-based semantical theories), then one makes the semantical standpoint sound inadequate for linguistic analysis through formalised description and reasoning mechanisms, given the inadequacy of a holistic approach to an enterprise of that sort.

Now, what seems remarkable about Lorenzen's and Hintikka's game-based approach to the characterisation of logical connectives is that a (semi-)formal account of Wittgenstein's "the *use* of the logical sign in the language-game" for the case of the language of mathematics is given, and yet no prohibitively insurmountable difficulty is dealt with by handling language in a manageably holistic fashion. It actually makes the explanation of use of logical constants 'atomistic', in the sense that each individual connective is treated separately, and 'functional' in the sense that each connective gets its meaning from the explanation of the 'rules of the game'. Moreover, each connective is explained by stating in each case what can be deduced from the propositions with those logical signs as the major connective, and the game follows after the *elimination* step is made. In other words, the rules are laid down by means of explanations of what (immediate) consequences one is allowed to draw from the corresponding propositions. It is then left implicit, (as in the context of a proper game where moves other than those stated are simply invalid, nonsensical), that any consequence beyond the one(s) indicated cannot be drawn from the proposition(s), whatever context they may be uttered in (wherever the game is played!). And this seems to be a very reasonable rationale to be followed by any theory of meaning which, whereas taking into account contexts (language-games) as frames of reference, is meant to be placing the semantical emphasis on the use and functionality of logical signs ("All signals get their sense from circumstances, it is true; but the way they function in the circumstances is different."[24]).

[24]p. 76 of Geach, P. T. (ed.): 1988, *Wittgenstein's Lectures on Philosophical Psychology 1946–47*, (Notes by P. T. Geach, K. J. Shah & A. C. Jackson), Harvester-Wheatsheaf, Hemel Hempstead, xv+348pp.

As a matter of fact, *use* (as *usefulness*) is not just a concept arbitrarily chosen by Wittgenstein as the basis for his semantical investigations, for it entails the crucial concept of *intentionality*. And, as a matter of course, there is a great deal of *intentional* ingredients in Wittgenstein's concept of meaning as use. That is what makes his semantic theory fundamentally different from many of the previous accounts of the relation between language and the world, which were mostly extensional and followed Augustine's tradition "as I heard words repeatedly used in their proper places in various sentences, I gradually learnt to understand what objects they signified". In such an *intentional* view of language as Wittgenstein's, there is no place for interpretations as endpoints for settling meaning. Moreover, in the light of such an alternative account of the links between language and reality, one can say with Wittgenstein that the phrase "the interpretation of signs" is misleading, and that one ought to say "the use of signs" instead[25], because the former suggests that a sign is to be understood by perhaps another sign which does not need any further interpretation, whereas the latter conveys a much broader conceptual basis which can accommodate intentionality: the kind of 'manageable' holism, which we are here referring to as 'functional atomism' to emphasise that the words of a language are identified as individuals which have the meaning determined by their function/use/rôle/usefulness in the calculus of language.

In the *Investigations* Wittgenstein speaks of 'interpretations hanging in the air along with what they interpret':

> "But how can a rule shew me what I have to do at *this* point? Whatever I do is, on some interpretation in accord with the rule."—That is not what we ought to say, but rather: any interpretation still hangs in the air along with what it interprets, and cannot give it any support. Interpretations by themselves do not determine meaning.[26]

The pictorial device of 'hanging in the air' is sometimes replaced by the figure of a 'rider' which stands there to help giving the sentence its meaning. This is the particular presentation of the shortcomings of interpretations as semantical devices that appears as early as in *Philosophical Grammar*, which collects writings from the early thirties:

> But an *interpretation* is something that is given in signs. It is this interpretation as opposed to a different one (running differently).—So when we wanted to say "Any sentence still stands in need of an

[25] §32, Part III, p. 70 of the *Philosophical Remarks*.

[26] §198, Part I, p. 80^{e} of the *Investigations*, op. cit..

> interpretation", that meant: no sentence can be understood without a rider.[27]

The 'rider' of a sentence can only disappear when the element of *intentionality* is acknowledged, and the intention is uncovered by a demonstration of the conclusions one can draw from the sentence. The intentional aspect of a sentence is not made clear until one makes explicit where one intends to arrive at by asserting it. Compare, e.g., 'what do you mean?' with 'what are you trying to say?', 'what conclusion(s) do you want one to draw from what you have just said?', 'what is your intention when asserting the sentence?', etc.. And it is 'intention' which is going to play a crucial rôle in breaking the apparently infinite chain of interpretations. And indeed, in another remark from the same *Philosophical Grammar*, he expresses the view that *intention* can be a useful concept in the breaking of such a chain of interpretations, but does not sound too satisfied when saying that such a concept is of a 'psychological' rather than 'logical' nature:

> By "intention" I mean here what uses a sign in a thought. The intention seems to interpret, to give the final interpretation; which is not a further sign or picture, but something else—the thing that cannot be further interpreted. But what we have reached is a psychological, not a logical terminus.[28]

In attempting to extract from Wittgenstein's early as well as later writings a certain uniformity with respect to the connections between the meaning of a proposition and the (immediate) consequences one is allowed to draw from it, it is also our endeavour to demonstrate that a logical counterpart to such a 'psychological' terminus, namely *intention*, is already present in his semantic theory, in spite of an apparent lack of the explicit and definite acknowledgement that this is the case by Wittgenstein himself.

4. Meaning and the explanation of consequences

By contrast with the wide popularity enjoyed by the connections between *meaning* and *use* that Wittgenstein's (especially later) writings convey, the position on the connections between the *meaning* of an assertion and the explanation of (immediate) *consequences* one is allowed to draw from it is hardly emphasised in most of the relevant literature on Wittgenstein's

[27] §9, p. 47 of *Philosophical Grammar*, Part I.

[28] Ibid., §98, p. 145.

semantic theory. Nevertheless, we find that without proper attention to this particular aspect of his writings on the notion of meaning, the understanding of Wittgenstein's use-based semantic theory must remain incomplete. There are a great many explicit references to such a crucial semantical standpoint especially, as we have said previously, throughout the writings of his later phase.

For example, in a typescript dictated in late autumn 1947 (TS 229), whose underlying manuscripts stem from the period between May 10th 1946 and October 11th 1947, published as *Remarks on the Philosophy of Psychology*, Vol. I, one can find a remark relating what an utterance means and what one is permitted to draw from it (our **emphasis**):

> ... someone who does philosophy or psychology will perhaps say "*I* feel that I think in my head". But what that means he won't be able to say. For he will not be able to say *what* kind of feeling that is; but merely to use the expression that he 'feels'; as if he were saying "*I* feel this stitch *here*". Thus he is unaware that **it remains to be investigated what his expression "I feel" means here, that is to say: what consequences** [*Konsequenzen*] **we are permitted to draw from this utterance.** Whether we may draw the same ones as we would from the utterance "I feel a stitch here".[29]

Here he uses the German '*Konsequenz*', as opposed to '*Folge*' which is used more frequently in his attempts to relate the consequences one can draw from an assertion and its meaning.

From another typescript dictated in early autumn 1948 (TS 232), bearing remarks which stem from manuscripts written between November 1947 and August 25th 1948, also published as *Remarks on the Philosophy of Psychology*, but this time Vol. II, one can see remarkably direct and explicit references to the connections between the meaning of a statement and what can be deduced from it.

> For the question is not, 'What am I doing when . . .?' (for this could only be a psychological question) – but rather, 'What meaning does the statement have, what can be deduced from it, *what consequences does it have*?'[30]

In other notebooks from a particularly late period of his later phase, namely the two last years of his life, which are collected in the book published

[29] §350, p. 69[e] of the *Remarks on the Philosophy of Psychology*, Vol. I.

[30] §38, p. 8[e] of the *Remarks on the Philosophy of Psychology*, Vol. II.

under the title of *On Certainty* as a direct reference to the main subject of the remarks, one can find clear traces of what one is tempted to call one of the canons of L. Wittgenstein's philosophical writings. An answer to the question 'what follows from your statement' plays the rôle of settling its meaning, giving its final interpretation, making the intention explicit (our **emphasis**):

> I KNOW that this is my foot. I could not accept any experience as proof to the contrary.—That may be an exclamation; **but what** *follows* **from it?** At least that I shall act with a certainty that knows no doubt, in accordance with my belief.[31]

Another passage from the same *On Certainty*, this time dated in the original as 27.3.51, again reflects his position with respect to the fundamental rôle of the demonstration of the consequences of an assertion in settling its meaning. It says (our **emphasis**):

> One is often bewitched by a word. For example, by the word "know". Is God bound by our knowledge? Are a lot of our statements *incapable* of falsehood? For that is what we want to say.
> I am inclined to say: "That *cannot* be false." That is interesting; **but what consequences has it?**[32]

5. Use and the explanation of consequences

Similarly to the previous case, it does not seem to be widely acknowledged the attempt Wittgenstein has made in demonstrating the close links between the *use* of a proposition and the *consequences* one can draw from it. Nevertheless, one can demonstrate that such an attempt is made very clear in later writings, by contrast with the case of *meaning* and *consequences*, which although less emphasised in the literature, yet can be traced back in early remarks as we will see in the sequel.

From the notes of a series of *Lectures on Religious Belief*, given in Cambridge sometime around the summer of 1938, taken by Y. Smythies and published in *Lectures and Conversations on Aesthetics, Psychology and Religious Belief*, one can choose a passage where the links between *use* and *consequences* are made very explicit (our **emphasis**):

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[31] §360, p. 47e of *On Certainty/ Über Gewissheit*.

[3216] Ibid., §§435–7, p. 57e.

> Yes, this might be a disagreement— **if he himself were to use the word in a way in which I did not expect, or were to draw conclusions I did not expect him to draw**.[33]

In the first volume of the *Last Writings on the Philosophy of Psychology*, which is not based on a typescript as is the case for the previously published *Remarks on the Philosophy of Psychology*, but on manuscript writings dating from the period between 22 October 1948 and 22 March 1949 except for the last remark dated 20 May (second half of MS 137 and the whole of MS 138), one can find examples of very explicit references to the connections between the use of words and the consequences of the corresponding utterance. Presented sometimes in a positive way such as in (our **emphasis**):

> What are you telling me when you *use* the words . . .? What can I do with this utterance? **What consequences does it have?**[34]

and in (our **emphasis**):

> What does anyone tell me by saying "Now I see it as . . ."? **What consequences has this information?** What can I do with it?[35]

(which also appears in the *Investigations*, Part II, section xi, p. 202^{e}) or in a negative manner such as in (our **emphasis**):

> The report "The word . . . was crammed full of its meaning" **is used quite differently, has quite different consequences**, from "It had the meaning . . .".[36]

his remarks on the connections between the use of a sentence and the consequences one is allowed to draw from it constitute a key ingredient in his investigations on the concept of meaning.

Considering that his main focus of attention in the later phase is indeed on the 'logic of ordinary language', rather than on the technicalities of Frege-Russell's logical calculus, it is not quite surprising that Wittgenstein's very late writings on meaning and use, strikes one as so intuitively appealing. By contrast, in early as well as not-so-late writings, in spite of a clear attempt

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[33] p. 71 of the *Lectures and Conversations on Aesthetics, Psychology and Religious Belief*.

[34] §624, p. 80^{e} of *Last Writings I*, op. cit..

[35] Id. Ibid., §630.

[36] Ibid., §785, p. 100^{e}.

to show that 'meaning' is better understood as 'use', generally speaking there is no clear-cut, incisive account of what should be taken as the logical counterpart of the latter. There are, however, a few references to meaning-use-consequences scattered amongst early writings, as we have attempted to demonstrate in a previous opportunity.[37] Unfortunately, the remaining general picture of the transitional period suggests a certain hesitation which is revealed quite clearly when the rôle of proofs in mathematics is analysed. In the writings of the early period of his later phase, documented in the notebooks containing remarks mainly on the foundations of mathematics now published as *Philosophical Remarks* and *Philosophical Grammar*, there is a clear shift in the focus of attention from truth-values and truth-functions which were characteristic of the *Tractatus*, to mathematical proofs. The concept of mathematical proof is thoroughly investigated, and in some passages it is even placed in a rather privileged position with respect to the semantics of the mathematical language. One of the lessons taught by those 'transitional' writings seems to be that (using Wittgenstein's own words) "when one wants to know the meaning, the sense, of a proposition, one has to look at what its proof proves".

It does not take too long, however, for a revision of the position on the connections between meaning and proofs to be made, and it is in the *Remarks on the Foundations of Mathematics*, written between 1941–44, that the revision is made explicit:

> I once said: 'If you want to know what a mathematical proposition says, look at what its proof proves'.[38] Now is there not both truth and falsehood in this? For is the sense, the point, of a mathematical proposition really clear as soon as we can follow the proof?[39]

A fair number of remarks which reveal a much less explicit statement of the revision made with respect to the connections between the meaning of a proposition and its proof appear in later stages of the later period. One of them, from the same *Remarks on the Foundations of Mathematics* and written around 1943/1944, says:

> The proof of a proposition certainly does not mention, certainly does not describe, the whole system of calculation that stands behind the proposition and gives it its sense.[40]

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[37] Cf. [6].

[38] Cf. *Philosophical Grammar*, p. 369; *Philosophical Remarks*, op. cit., pp. 183–4.

[39] §10, Part VII, p. 367 of the *Remarks on the Foundations of Mathematics*.

[40] Ibid., §11, Part VI, p. 313.

which reveals his concern on the insufficiency of the explanation of what counts as a proof (i.e. explanation of assertability conditions) with respect to the explanation of meaning. The apparently unworkable holism is shown perfectly feasible to the formulation of a theory of meaning for the language of mathematics by his suggestions on the connections between the meaning of a proposition and the explanation of the immediate consequences one can draw from it:

> One learns the meaning [*Bedeutung*] of 'all' by learning that 'fa' follows from '$(x).fx$'.
>
> (...)
>
> And the meaning [*Sinn*] of '$(x).fx$' is made clear by our insisting on 'fa''s following from it.[41]

It is as if the parts of language are understood only in relation to other neighbouring parts, as D. Pears points out.[42]

His insistence on the idea that "the meaning of a word is determined by its rôle in the language" would be understood in the following way: by showing how to 'eliminate', 'get rid of' the logical sign, demonstrating the conclusions one can draw from a proposition which has it as its major connective, one is demonstrating the rôle it plays in the calculus/system of propositions.

Unfortunately, his acknowledgement of the inappropriateness of using mathematical proofs and assertability conditions as semantical notions does not seem to have been taken into consideration by some 'verificationistic' theories of meaning which still look for a conceptual basis in Wittgenstein's semantic theory and suggest that 'meaning is use' is their underlying semantical paradigm. Wittgenstein's clear emphasis on the *deductions* that can be made from a proposition as explanations of its meaning does not appear to have been given the attention it would seem to deserve by those who advocate a use-based semantic theory but still insist on the primacy of assertability conditions. Neither there seems to be much concern about this particular aspect of Wittgenstein's very late writings in most of the literature on Wittgenstein's philosophy, to the best of our knowledge.

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[41] Ibid., Part I, §10, p. 41 and §11, p. 42, respectively.

[42] "The new idea [of W.'s later theory of language] was that we experience each part of language in its relations to other, neighbouring parts.", p. 171 of [36].

6. Early signs of 'meaning – use/usefulness – consequences'

The intuition regarding the connections between meaning and use (as usefulness), which appears explicitly in very late remarks such as the one shown in the opening lines as well as in the title of this paper, and underlies the "words are tools"[43] conceptual link made mainly in Part I of the *Investigations*, seems to find its roots in very early remarks documented in early writings such as the *Notebooks* and the *Tractatus*. If the former shows the early signs of the connections between meaning and use — "The way in which language signifies is mirrored in its use"[44] —, in the latter one can find remarks such as:

> If a sign is *useless*, it is meaningless. That is the point of Occam's maxim.[45]

and 'useless' here is to be understood as 'purposeless', as it becomes clear later in the same *Tractatus*:

> Occam's maxim is, of course, not an arbitrary rule, not one that is justified by its success in practice: its point is that *unnecessary* units in a sign-language mean nothing.
> Signs that serve *one* purpose are logically equivalent, and signs that serve *none* are logically meaningless.[46]

In the former, the question whether the logical constants exist is not even considered to be relevant, given that they can 'disappear'. And again, the position can be seen to be very reasonable and coherent with the one expressed in the remark above if one looks at such 'constants' as devices, tools, which find meaning in their *use/usefulness*. In this particular respect, Wittgenstein's words in the *Notebooks* are put in the following way:

> With the logical constants one need never ask whether they exist, for they can even *vanish*![47]

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[43] "Think of the tools in a tool-box: there is a hammer, pliers, a saw, a screw-driver, a rule, a glue-pot, glue, nails and screws.—The functions of words are as diverse as the functions of these objects. (And in both cases there are similarities.)", §11, Part I, p. 6^{e} of the *Investigations*, op. cit..

[44] dated 11.9.16, p. 82e of the *Notebooks 1914-1916*.

[45] §3.328, p. 16 of the *Tractatus Logico-Philosophicus*.

[46] Ibid., §5.47321, p. 48.

[47] dated 25.10.14, p. 19^{e} of the *Notebooks*, op. cit..

In the same *Notebooks* the (introduction of the) sign '0' is suggested to obtain its meaning from its usefulness in making the decimal notation possible:

> Einführung des Zeichens ,,0" um die Dezimalnotation möglich zu machen: Die logische Bedeutung dieses Vorgehens.[48]

(translated as "The introduction of the sign "0" in order to make the decimal notation possible: the logical significance of this procedure.")

If meaning is to be understood as use/usefulness, logical connectives should be better seen as *instruments*, rather than as *constants*, and that is what seems to be suggested by Wittgenstein even in the *Tractatus* where the truth-values based semantical view inherited from Frege and Russell characterises the 'early Wittgenstein' as distinct from the 'later Wittgenstein and the use-based semantical view'. And one can see his rather original point of view on logic and language *versus* reality, when he suggests that logical connectives are *tools*, therefore the idea that 'logical connectives are *constants*' and the viewpoint that 'there are logical constants, namely "$\vee$", "$\supset$", "." ("$\wedge$"), etc.' are hardly tenable. He makes the point by recalling the so-called interdefinability of such 'constants':

> At this point it becomes manifest that there are no 'logical objects' or 'logical constants' (in Frege's and Russell's sense).[49]

Under such a viewpoint, it seems rather clear that insofar as being understood as useful *instruments*, the logical connectives would not have to be looked at as 'primitive' signs. This appears to be suggested by Wittgenstein in (again the *Tractatus*):

> It is self-evident that $\vee$, $\supset$, etc. are not relations in the sense in which right and left etc. are relations.
> The interdefinability of Frege's and Russell's 'primitive signs' of logic is enough to show that they are not primitive signs, still less signs for relations.[50]

Furthermore, in spite of resorting to truth-values and truth-functions when providing a precise account of meaning for logical propositions by

[48] dated 16.11.14, p. 31 of the *Notebooks*, op. cit..

[49] §5.4, p. 44 of the *Tractatus*, op. cit..

[50] Ibid., §5.42, p. 44.

means of his "a proposition is a truth-function of elementary propositions. (an elementary proposition is a truth-function of itself.),"[51] Wittgenstein did not seem to be too convinced that the concept of truth-function could work equally well for all logical concepts and connectives of Frege-Russell's logical calculus. For example, the generality concept in which the universal quantifier was embedded was explicitly dissociated from truth-functions in:

> I dissociate the concept *all* from truth-functions.
> Frege and Russell introduced generality in association with logical product or logical sum. This made it difficult to understand the propositions '$(\exists x).fx$' and '$(x).fx$', in which both ideas are embedded.[52]

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[51] Ibid., §5, p. 36.

[52] Ibid., §5.521, p. 51. The use of 'dissociate' in the translation by D. Pears and B. McGuinness appears to reinforce Wittgenstein's deliberate intention in "Ich trenne den Begriff *Alle* von der Wahrheitsfunktion" to disconnect one from another. This is clear from the correspondence between Wittgenstein and the first (English) translator C. K. Ogden. In the 'Comments II on Separate Sheets', from *Letters to C. K. Ogden*, p. 61, he says:

> *5.521* I mean "separate" and *not* "derive". They were connected and I separate them.

and the comments by the Editor, p. 65:

> *5.521* The passage under discussion runs: "I separate the concept *all* from the truth-function."
>
> Ogden had queried whether "separate" should not be changed to "derive".

From this it should be clear that it is unreasonable to say that Wittgenstein in the *Tractatus* "construes quantified sentences as conjunctions and disjunctions" (p. 110 of [30], *Investigating Wittgenstein*). Hintikka & Hintikka's conclusion seems to draw on Moore's saying that Wittgenstein in 1930–33, "went on to say that one great mistake which he made in the *Tractatus* was that of supposing that in the case of *all* classes "defined by grammar", general propositions were identical with logical products or with logical sums (meaning by this logical products or sums of the propositions which are values of fx) as, according to him, they really are in the case of the class of "primary colours". He said that, when he wrote the *Tractatus*, he had supposed that *all* such general propositions were "truth-functions"; (...) He said that, when he wrote the *Tractatus*, he would have defended the mistaken view which he then took by asking the question: How can $(x).fx$ possibly entail fa, if $(x).fx$ is not a logical product? And he said that the answer to this question is that where $(x).fx$ is not a logical product, the proposition "$(x).fx$ entails fa" is "taken as a primary proposition", whereas it is a logical product this proposition is deduced from other primary propositions.", (our emphasis), p. 3 of Moore, G. E.: 1955, 'Wittgenstein's Lectures in 1930–33', in *Mind*, vol. 54, n. 253, pp. 1–27. Wittgenstein might well have said that, given that he was very critical, almost ruthless, with himself and he often insisted on the big mistakes he made in the *Tractatus*. For example, he says in a section entitled '*Criticism of my former view of generality*' of the *Philosophical Grammar*, p. 268, that his "view about general propositions was that $(\exists x).\varphi x$ is a logical sum". Nonetheless, nowhere in the *Tractatus* the existential quantifier

For the particular case of the logical notion of generality and the corresponding connective for universal quantification, the notion of 'following from' was preferred to the more *Tractatus*-characteristic truth-function:

> (Daß man aus (x).fx auf fa schließen kann, das zeigt, daß die Allgemeinheit auch im Symbol ≫(x).fx≪ vorhanden ist.)

(translated as 'The possibility of inference from $(x).fx$ to fa shows that the symbol $(x).fx$ itself has generality in it.'[53])

The inference demonstrating the (immediate) consequences one can draw from the proposition '$(x).fx$' is actually making clear that the distinguishing characteristic of the major logical sign (namely '$(x).$', *for all*) is that of generality. From this standpoint, one can read many of the remarks contained in the *Tractatus* itself in a way which has, to the best of our knowledge, hardly been done in the literature on the early Wittgenstein, and place them still in line with the remarks from the later Wittgenstein using the same connections between *meaning*, *use/usefulness* and *consequences*. For example, there is a remark concerning the view that in logic the performance of deductions ('*Folgern*') happen *a priori*, and it seems that one can understand this in the light of, e.g., Hintikka's *semantical games*, a (semi-)formal counterpart to later Wittgenstein's 'the meaning of a sign is its use in the language-game', where the logical connectives are given meaning by means of rules of 'one-step' deduction. And indeed, if "before a proposition can have a sense, it must be completely settled what propositions follow from it"[54], according to the *Prototractatus*, then:

> Alles Folgern geschieht a priori.

(translated as 'All deductions are made a priori.'[55]) according to the *Tractatus*. It does not seem unreasonable to say that these remarks can be understood along the same lines advocated here by simply acknowledging that to

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is treated as a logical sum. Moreover, as we shall see below, early remarks bear witness of the different, 'non-truth-functional' treatment he gave to the universal quantifier.

[53] Ibid., §5.1311, p. 38.

[54] §3.20103, p. 65 of the *Prototractatus*. An almost identical remark already appears in the *Notebooks 1914–16*: "We might demand definiteness in this way too: if a proposition is to make sense then the syntactical employment of each of its parts must be settled in advance.—It is, e.g., not possible *only subsequently to come upon* the fact that a proposition follows from it. But, e.g., **what propositions follow from a proposition must be completely settled before that proposition can have a sense!**", *Notebooks*, dated 18.6.15, p. 64[e]. (Our **emphasis**)

[55] §5.133, p. 39, of the *Tractatus*, op. cit..

show the consequences which are allowed to be drawn from a proposition is to demonstrate what can be done with it, the deductions happening *a priori* as part of the 'definition'. Moreover, in the light of this same conceptual basis one has to say that (using early Wittgenstein's own words), "logic must look after itself"[56] just because for a proposition to be defined and have a sense in logic, the rule showing the consequences (deductions) one can draw from the proposition must be given *a priori*. Therefore, "we cannot give a sign the wrong sense"[57], and so, the (one-step) 'deductions' are made *a priori* simply because they *determine* the meaning of the proposition.

Here it is relevant to reemphasise that one can find examples of remarks suggesting the links between the meaning of an assertion and what can be deduced from it, even in very early writings, such as: "A proposition affirms every proposition that follows from it."[58], and: " '$p.q$' is one of the propositions that affirm 'p' and at the same time one of the propositions that affirm 'q'."[59] of the *Tractatus*, and, e.g.:

> That the *proposition* "ϕa" can be inferred from the *proposition* "(x).ϕx" shows how generality is present even in the *sign* "(x).ϕx".[60]

of the *Notebooks*.

As early as 1912, in a letter to Russell, dated 1.7.12, he had already made a remark which is essentially similar to the one above (our **emphasis**):

> Will you think that I have gone mad if I make the following suggestion?: The sign $(x).\varphi x$ is not a complete symbol but has meaning only in an inference of the kind: **from** $\vdash \varphi x \supset_x \psi x.\varphi(a)$ **follows** $\psi(a)$. Or more generally: **from** $\vdash (x).\varphi x.\varepsilon_0(a)$ **follows** $\varphi(a)$. I am—of course—most uncertain about the matter but something of the sort might really be true.[61]

7. Concluding remarks

By attempting to fill in the two lesser known sides of the semantic triangle-like diagram with vertices in *meaning*, *use* and (the explanation of

[56]Ibid., §5.473, p. 47.

[57]Id. Ibid., §5.4732.

[58]Ibid., §5.124, p. 38.

[59]Id. Ibid., §5.1241.

[60]dated 24.11.14, p. 32^{e} of the *Notebooks*, op. cit..

[61]Letter *R.3*, p. 12 of Wittgenstein, L.: 1974, *Letters to Russell, Keynes and Moore*, Ed. with an Introd. by G. H. von Wright, (assisted by B. F. McGuinness), Basil Blackwell, Oxford, 190pp.

the immediate) *consequences* we have tried to demonstrate how one finds a formal counterpart to Wittgenstein's 'meaning-is-use' in the proof-theoretic semantics based on convertibility (normalisation). We had previously presented a reformulated type theory and its underlying 'semantics of use' [8], demonstrating its appropriateness in studying the logical foundations of computation.

In [11] we have pointed out the parallels between our proposal and other approaches to the semantics of logical connectives based on the explanation of the (immediate) consequences of the corresponding proposition, such as Lorenzen's [32] *dialogical games* [32, p. 25] and Hintikka's *semantical games* [28, p. 156]; [29, p. 3]. The general underlying principle is that of the logical principle of inversion, uncovered by Gentzen, and later by Lorenzen [31]: the elimination procedure is the exact inverse of the introduction, therefore all that can be asked from an assertion is what is indicated in the explanation of elimination procedure. One can look at Lorenzen's dialogical games and try to show where the principle of inversion resides in the game-approach by comparing it to the normalisation-approach (read '$\triangleright_\beta$' as 'β-normalises to'; for more details of the formal rules, the reader is referred to the appendix):

Assertion/*Introd.* (Myself/	Attack/*Elim.* Nature/	$\triangleright_\beta$	Defense Game continues)
Conjunction ('$\wedge$'):			
$A_1 \wedge A_2$	$L?$		A_1
$A_1 \wedge A_2$	$R?$		A_2
$\langle a_1, a_2\rangle : A_1 \wedge A_2$	$\mathtt{FST}(\langle a_1, a_2\rangle)$	$\triangleright_\beta$	$a_1 : A_1$
$\langle a_1, a_2\rangle : A_1 \wedge A_2$	$\mathtt{SND}(\langle a_1, a_2\rangle)$	$\triangleright_\beta$	$a_2 : A_2$
Disjunction ('$\vee$'):			
$A_1 \vee A_2$	?		A_1
$A_1 \vee A_2$	?		A_2
$\mathtt{inl}(a_1) : A_1 \vee A_2$	$\mathtt{CASE}(\mathtt{inl}(a_1), s'_1 d(s_1), s'_2 e(s_2))$	$\triangleright_\beta$	$d(a_1/s_1) : C$
$\mathtt{inr}(a_2) : A_1 \vee A_2$	$\mathtt{CASE}(\mathtt{inr}(a_2), s'_1 d(s_1), s'_2 e(s_2))$	$\triangleright_\beta$	$e(a_2/s_2) : C$
If-then ('$\rightarrow$'):			
$A \rightarrow B$	A ?		B
$\lambda x.b(x)$	$\mathtt{APP}(\lambda x.b(x), a)$	$\triangleright_\beta$	$b(a/x) : B$
Universal Quantifier ('$\forall$'):			
$\forall x^D.F(x)$	$a : D$?		$F(a)$
$\Lambda x.f(x)$	$\mathtt{EXTR}(\Lambda x.f(x), a)$	$\triangleright_\beta$	$f(a/x) : F(a)$

Existential Quantifier ('$\exists$'):

$\exists x^D.F(x)$	?	$t : D, F(t)$
$\varepsilon x.(f(x), a)$	$\mathtt{INST}(\varepsilon x.(f(x), a), \acute{g}\acute{t}d(g, t)) \quad \triangleright_\beta$	$d(f/g, a/t) : C$

Propositional Equality ('$\doteq$'):

$\doteq_D (a, b)$	?	$a =_t b : D$
$r(a, b)$	$\mathtt{REWR}(r(a, b), \acute{t}d(t)) \quad \triangleright_\beta$	$d(r/t) : C$

If we look at the conclusion of *reduction* inference rules, and we take the `DESTRUCTOR` as being the Attack (or 'Nature', its counterpart in the terminology of Hintikka's Game-Theoretical Semantics), and the `constructor` as being the Assertion (or Hintikka's 'Myself'), it is not hard to see that the game-theoretic explanations of logical connectives find a counterpart in the functional interpretation with the semantics of convertibility [5]:

1. Whilst in both $\mathtt{FST}(\langle a_1, a_2\rangle)$ and $\mathtt{SND}(\langle a_1, a_2\rangle)$ the destructors allow access to either of the conjuncts, in $\mathtt{CASE}(\mathtt{inl}(a_1), \acute{s_1}d(s_1), \acute{s_2}e(s_2))$ and $\mathtt{CASE}(\mathtt{inr}(a_2), \acute{s_1}d(s_1), \acute{s_2}e(s_2))$ the destructor has no choice but to go to whichever disjunct comes from the introduction.
2. Whilst in $\mathtt{EXTR}(\Lambda x.f(x), a)$ the destructor has the choice of the witness a, in $\mathtt{INST}(\varepsilon x.(f(x), a), \acute{g}\acute{t}d(g, t))$ has no choice but to eliminate over the witness a which is inside the term $\varepsilon x.(f(x), a)$ built with the constructor because it was chosen by the introduction.
3. In $\mathtt{REWR}(r(a, b), \acute{t}d(t))$ the destructor does not have the choice as to the 'reason', since r will have been chosen by the time of the application of the introduction rule.

It would not seem unreasonable to say that the parallels mentioned above are to be attributed to the 'universal validity' of the underlying logical principle of inversion. (Recall that the cut-rule, sometimes said to be a fundamental rule for any calculus to have in order to be considered a logic (cf. [26]), can be seen as an instance of the principle of inversion.) The principle of inversion and the 'naturally' opposite sides appear in many different foundational frameworks:

- Proof Theory (Gentzen): Introduction / Elimination
- λ-Calculus (Church): Canonical Terms / Noncanonical Terms
- Dialogue Games (Lorenzen): Assertion / Attack

- Semantical Games (Hintikka): Myself / Nature

Concerning the use of such a framework for the development of a new foundational basis for proof-theoretic semantics, we have been engaged in the development of what we call *Labelled Natural Deduction* [17], which provides the basis for:

1. the study of various notions of implication by taking β-normalisation on $\rightarrow$-types as the main semantical device, and varying disciplines of assumption discharge and corresponding λ-abstractions [23, 24];
2. the reassessment of Skolem and Herbrand fundamental results, as well as the various formulations of the notion of singular terms (including Hilbert's ε-calculus) via the functional interpretation [15];
3. the study of modal logics via the use of an extension of the variable-binding technique to deal with arbitrary 'structured' collections of formulas and domains ('worlds'): the procedure is similar to the one followed in (1) where the β-normalisation on terms containing 'world-variables' is kept fixed, while the disciplines of discharging assumptions of the form 'let $\mathbb{S}$ name an arbitrary world' (and corresponding Λ-abstraction disciplines on terms containing $\mathbb{S}$) are modified to suit the particular kind of necessity [16].

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Appendix: Normalisation and the explanation of the consequences

An important step in the characterisation of the logical connectives which is not covered by the (language-based) intuitionistic account of meaning based on proofs is the explanation of the functionality of the logical sign in the calculus. In the formal apparatus, this means that the explanation as to how the `DESTRUCTORS` operate on terms built up by the `constructors`, i.e. the explanation of what (immediate) consequences one can draw from the corresponding proposition, does not play a major semantical rôle for the intuitionists.[62] Within the functional interpretation, which we believe might have been implicit in Tait's semantics of convertibility [45], this aspect is given by the so-called β-normalisation rules. They have the rôle of spelling out the effect of an *elimination* inference on the result of *introduction* steps. So, the explanation of the normalisation of noncanonical proofs, i.e. those which contain 'redundant' steps identified by an *introduction* inference immediately followed by an *elimination* inference, are framed in the following way (where '$\triangleright_\beta$' represents 'β-converts/normalises to'):

$\wedge$-*reduction*

$$\dfrac{\dfrac{a_1 : A_1 \qquad a_2 : A_2}{\langle a_1, a_2\rangle : A_1 \wedge A_2}\,\wedge\text{-}intr}{\mathtt{FST}(\langle a_1, a_2\rangle) : A_1}\,\wedge\text{-}elim \qquad \triangleright_\beta \qquad a_1 : A_1$$

$$\dfrac{\dfrac{a_1 : A_1 \qquad a_2 : A_2}{\langle a_1, a_2\rangle : A_1 \wedge A_2}\,\wedge\text{-}intr}{\mathtt{SND}(\langle a_1, a_2\rangle) : A_2}\,\wedge\text{-}elim \qquad \triangleright_\beta \qquad a_2 : A_2$$

[62] In [9] we have argued that the *introduction* rules only cover one aspect, namely the grammatical (formational) aspect: they only say how to construct a proof, leaving the aspect as to how to de-construct (challenge) this same proof untouched.

$\vee$-*reduction*

$$\dfrac{\dfrac{a_1 : A_1}{\mathtt{inl}(a_1) : A_1 \vee A_2}\vee\text{-}intr \qquad \begin{matrix}[s_1 : A_1] \\ d(s_1) : C\end{matrix} \qquad \begin{matrix}[s_2 : A_2] \\ e(s_2) : C\end{matrix}}{\mathtt{CASE}(\mathtt{inl}(a_1), \acute{s_1}d(s_1), \acute{s_2}e(s_2)) : C}\vee\text{-}elim \quad \triangleright_\beta \quad \begin{matrix}[a_1 : A_1] \\ d(a_1/s_1) : C\end{matrix}$$

$$\dfrac{\dfrac{a_2 : A_2}{\mathtt{inr}(a_2) : A_1 \vee A_2}\vee\text{-}intr \qquad \begin{matrix}[s_1 : A_1] \\ d(s_1) : C\end{matrix} \qquad \begin{matrix}[s_2 : A_2] \\ e(s_2) : C\end{matrix}}{\mathtt{CASE}(\mathtt{inr}(a_2), \acute{s_1}d(s_1), \acute{s_2}e(s_2)) : C}\vee\text{-}elim \quad \triangleright_\beta \quad \begin{matrix}[a_2 : A_2]\downarrow \\ e(a_2/s_2) : C\end{matrix}$$

where '$\acute{}$' is an abstractor which forms course-of-values terms such as '$\acute{x}d(x)$' where 'x' is bound, and discharges the corresponding assumption.[63]

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[63]The device of variable-binding, and the idea of having terms representing incomplete 'objects' whenever they contain free variables, were both introduced in a systematic way by Frege in his *Grundgesetze*. As early as 1893 Frege developed in his *Grundgesetze I* what can be seen as the early origins of the notions of *abstraction* and *application*, when showing techniques for transforming functions (expressions with free variables) into course-of-values terms (expressions with no free variables) by means of an 'introductory' operator of abstraction producing the *Wertverlauf* expression, e.g., '$\acute{\varepsilon}f(\varepsilon)$', and the effect of its corresponding 'eliminatory' operator '$\cap$' on a *course-of-values* expression as in §3, p. 7. Cf. also §34, p. 52ff:

> "(...) it is a matter only of designating the value of the function $\Phi(\xi)$ for the argument Δ, i.e., $\Phi(\Delta)$, by means of "Δ" and "$\acute{\varepsilon}\Phi(\varepsilon)$". I do so in this way:
>
> "$\Delta \cap \acute{\varepsilon}\Phi(\varepsilon)$"
>
> which is to mean the same as "$\Phi(\Delta)$"."

(Note the similarity to the rule of functional *application*, where 'Δ' is the argument, '$\acute{\varepsilon}\Phi(\varepsilon)$' is the function, and '$\cap$' is the operator '`APP`'.)

Expressing how pleased he was with the introduction of a variable-binding device for the functional calculus (recall that the variable-binding device for the logical calculus had been introduced earlier in the *Begriffsschrift*), Frege says:

> "The introduction of a notation for courses-of-values seems to me to be one of the most important supplementations that I have made of my *Begriffsschrift* since my first publication on this subject." (*Grundgesetze I*, §9, p. 15f.)

The idea of forming *course-of-values* function-terms by abstracting from the corresponding free variable is in fact very useful in representing the handling of assumptions within a Natural Deduction style calculus. In particular, when the Natural Deduction presentation system is based on a 'labelling' mechanism the binding of free variables in the labels corresponds to the discharge of corresponding assumptions. In the sequel we shall be using 'abstractors' (such as 'λ' in '$\lambda x.f(x)$') to bind free-variables and discharge the assumption labelled by the corresponding variable.

$\rightarrow$-*reduction*

$$\dfrac{a : A \qquad \dfrac{\begin{array}{c}[x : A]\\ b(x) : B\end{array}}{\lambda x.b(x) : A \rightarrow B}\rightarrow\text{-}intr}{\mathtt{APP}(\lambda x.b(x), a) : B}\rightarrow\text{-}elim \qquad \triangleright_\beta \qquad \begin{array}{c}[a : A]\\ b(a/x) : B\end{array}$$

$\forall$-*reduction*

$$\dfrac{a : D \qquad \dfrac{\begin{array}{c}[x : D]\\ f(x) : F(x)\end{array}}{\Lambda x.f(x) : \forall x^D.F(x)}\forall\text{-}intr}{\mathtt{EXTR}(\Lambda x.f(x), a) : F(a)}\forall\text{-}elim \qquad \triangleright_\beta \qquad \begin{array}{c}[a : D]\\ f(a/x) : F(a)\end{array}$$

$\exists$-*reduction*

$$\dfrac{\dfrac{a : D \qquad f(a) : F(a)}{\varepsilon x.(f(x), a) : \exists x^D.F(x)}\exists\text{-}intr \qquad \begin{array}{c}[t : D, g(t) : F(t)]\\ d(g, t) : C\end{array}}{\mathtt{INST}(\varepsilon x.(f(x), a), \acute{g}\acute{t}d(g, t)) : C}\exists\text{-}elim \triangleright_\beta \qquad \begin{array}{c}[a : D, f(a) : F(a)]\\ d(f/g, a/t) : C\end{array}$$

where '$\acute{}$' is an abstractor which binds the free variables of the label, discharging the corresponding assumptions made in eliminating the existential quantifier, namely the 'Skolem'-type assumptions '$[t : D]$' and '$[g(t) : F(t)]$', forming the course-of-values term '$\acute{g}\acute{t}d(g, t)$' where both the Skolem-constant 't', and the Skolem-function 'g', are bound.

$\doteq$-*introduction*

$$\dfrac{a =_s b : D}{s(a, b) :\doteq_D (a, b)}$$

$\doteq$-*reduction*

$$\dfrac{\dfrac{a =_s b : D}{s(a, b) :\doteq_D (a, b)}\doteq\text{-}intr \qquad \begin{array}{c}[a =_t b : D]\\ d(t) : C\end{array}}{\mathtt{REWR}(s(a, b), \acute{t}d(t)) : C}\doteq\text{-}elim \qquad \triangleright_\beta \qquad \begin{array}{c}[a =_s b : D]\\ d(s/t) : C\end{array}$$

where '$\acute{}$' is an abstractor which binds the occurrences of the (new) variable 't' introduced with the local assumption '$[a =_t b : D]$' as a kind of 'Skolem'-type constant denoting the (presumed) 'reason' why 'a' was assumed to be equal to 'b'. (Recall the Skolem-type procedures of introducing new local assumptions in order to allow for the elimination of logical connectives where the notion of 'hiding' is crucial, e.g. disjunction and existential quantifier — in [15].)

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